



Acidified ZnCl_2 molten salt hydrate systems as hydrolytic media for cellulose I and II nanocrystal production: from rods to spheres

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Abstract Exploring environmentally benign and economic methods for the preparation of nanocelluloses with controllable morphology would promote their industrial production and utilization. Herein, cellulose I and II nanocrystals (CNC I and CNC II) of rod and sphere shapes were prepared directly from microcrystalline cellulose (MCC, cellulose I) via treatment by acidified zinc chloride molten salt hydrates. Upon adding a small amount of HCl (0.50/0.75/1.0 M) to ZnCl_2 (55/60/65%), the levels

of cellulose swelling and hydrolysis could be tuned. Rod-like CNCs I (length: 181.3–279.3 nm) were obtained with 55% ZnCl_2 . Rod-like CNCs with size reduction along with spherical CNCs were produced as ZnCl_2 concentration increased to 60%. Further raising the ZnCl_2 concentration to 65% resulted in CNC II with exclusively spherical shape (diameter: 25.8–292.3 nm). Under the same ZnCl_2 concentration, CNCs produced with high HCl concentration exhibited improved crystallinity (up to 68.4%) and thermal stability (T_{max} up to 343.5 °C). Furthermore, spherical CNC II displayed excellent suspension stability and emulsifying ability to stabilize oil/water Pickering emulsions. The best one stabilized 26% more emulsion volume than the commercially available CNCs and the emulsions showed shear and heat stability. Therefore, this study provides a sustainable method for production of CNCs with tailored characteristics and broad application potentials.

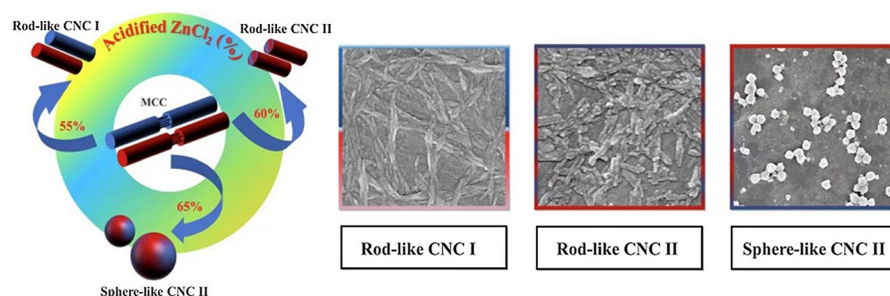
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Graphical abstract



Keywords Cellulose nanocrystals · Spherical nanocellulose · ZnCl_2 molten salt hydrate · Pickering emulsion · Cellulose I and II

Introduction

Cellulose nanocrystals (CNCs) are a type of highly crystalline nano-structured materials isolated from renewable and abundant lignocellulosic sources. They own a number of unique properties such as high Young modulus, high specific surface area, transparency, biodegradability and biocompatibility (Klemm et al. 2011; Moon et al. 2011) and thus are widely used for composite reinforcement, food additives, medical materials, adsorbents, energy storage materials, etc. (De France et al. 2021; Gu and Catchmark 2013; Hu et al. 2016; Klemm et al. 2018; Zhao et al. 2021). Acid hydrolysis of natural cellulose using mineral acids, such as sulfuric, hydrochloric, phosphoric acid and their mixtures (Moon et al. 2011), has been the most common method to selectively remove the less ordered regions of cellulose to yield CNCs. Relatively green solvents, such as ionic liquids (IL) (Pang et al. 2018) and deep eutectic solvents (DES) (Liu et al. 2017), are recently employed as efficient swelling and hydrolysis media to produce CNCs. Medium concentration, hydrolysis time and temperature have significant impacts on the yield and properties of the products. The resultant CNCs are generally stiff and rod-like with a length less than a micrometer and a width from several nanometers to tens of nanometers. These CNCs usually remain native cellulose I allomorph (CNC I) with parallel chain orientation as the original cellulose source. Only under restricted

reaction temperature and sulfuric acid concentration, cellulose II nanocrystals (CNC II) with anti-parallel fashion could be directly produced from cellulose I via a dissolution-regeneration process (Sebe et al. 2012; Xing et al. 2018, 2020). In most circumstances, CNC II is obtained from acid hydrolysis or oxidation of mercerized/regenerated cellulose II (Gong et al. 2017; Han et al. 2013; Yang et al. 2013).

Besides rod-like morphology, CNC II nanoparticles with spherical and granular shapes have been obtained from a variety of raw materials (Beaumont et al. 2016; Ye et al. 2018; Zhu et al. 2013). Typical production methods of spherical CNC II include ammonium persulfate (APS) oxidation (Zhu et al. 2013), 2,2,6,6-tetramethylpiperidine-1-oxyl (TEMPO) radical-mediated oxidation (Liu et al. 2020), carboxymethylation (Beaumont et al. 2016), and acid hydrolysis (Yan et al. 2015; Yang et al. 2013) of regenerated or mercerized cellulose. Most of spherical CNCs own diameters below 100 nm, only a few of them have average diameters in the range of 100–1000 nm (Tian et al. 2022).

Inorganic molten salt hydrates (MSH) are media that have a molar water-to-salt ratio approaching to the coordination number of the strongest hydrated cation. Thus the inner coordination sphere of the cation is tightly bound with the water molecules. A few MSH, such as $\text{ZnCl}_2 \cdot 3\text{H}_2\text{O}$, $\text{LiClO}_4 \cdot 3\text{H}_2\text{O}$ and $\text{LiI}_2 \cdot 2\text{H}_2\text{O}$, are able to dissolve cellulose by physical interaction with the polymer (Wang et al. 2016, 2020). MSH are inexpensive and easier to prepare compared to other cellulose-dissolving organic ionic liquids. Furthermore, they are environmentally benign without toxic and volatile organic compounds. The main component of MSH, the inorganic

salts, can be easily recovered and recycled for further use (Deng et al. 2015). Among different MSH systems, zinc chloride hydrate is frequently used due to its cheap price and ability to dissolve high molecular weight cellulose (Rodriguez Quiroz et al. 2019; Zhang et al. 2018). The cellulose-dissolving ability of the zinc chloride hydrates strongly depends on their hydrate composition ($\text{ZnCl}_2 \cdot R \text{H}_2\text{O}$, $R = 2\text{--}4$).

The utilization of acidified inorganic molten salt hydrates (AMSH) is demonstrated to promote cellulose hydrolysis by dissolving/swelling the polymer under mild reaction conditions (i.e. relatively low acid concentration, low temperature and short reaction time) (Cao et al. 1995). The inorganic salt in the reaction system dramatically increases the hydrogen activity and the salt-inorganic acid interaction raises the acidity of the media (Rodriguez Quiroz et al. 2019). Cellulose or lignocellulose can be effectively converted to pentose/hexose sugars in AMSH media (Deng et al. 2015). Very recently, CNC II was produced by a two-step method combining MSH pretreatment and acid hydrolysis/oxidation (Li et al. 2021; Liu et al. 2021). A mildly acidic lithium bromide trihydrate system (61% $\text{LiBr} + 2.5 \text{ mM H}_2\text{SO}_4$) and a zinc chloride trihydrate system (72% ZnCl_2) were used to pretreat bleached kraft pulp and microcrystalline cellulose (MCC), respectively, to obtain regenerated cellulose, which was further hydrolyzed or oxidized to generate CNC II. However, these studies did not investigate the impact of inorganic salt concentration on CNC structure.

In this work, we combined AMSH and sonication treatments to prepare CNCs with tailorable morphology and allomorph. We hypothesized that by controlling the penetration depth of the MSH to cellulose crystallites and the acidity of the system, the cellulose hydrolysis rate can be adjusted and CNCs with high crystallinity could be produced. Hydrochloric acid acidified zinc chloride molten salt hydrate systems were developed for facile and green isolation of CNCs from MCC. The ZnCl_2 and HCl concentrations varied to adjust the swelling degree and thus the hydrolysis rate of the cellulose. CNCs with tunable shapes (rod and sphere), sizes and allomorphs (cellulose I and II) are produced and systematically characterized in terms of morphology, particle dimensions, surface charge, chemical and crystalline structures, and thermal properties. To demonstrate the unique properties of the resulting spherical CNC II, they are

utilized as stabilizer for oil-in-water Pickering emulsions. Their emulsifying ability was compared to that of commercialized sulfuric acid prepared CNCs.

Experimental section

Materials

Microcrystalline cellulose (MCC) (AR 99%, DP 168) was purchased from Aladdin. The MCC particles were about 1–10 μm in diameter and 15–40 μm long (Fig. S1). Hydrochloric acid (36–38%, Guangzhou Chemical Reagent Factory), zinc chloride (ZnCl_2 , AR, 98%, Macleans) and Tween 20 (Tianjin Fuchen Chemical Reagents Factory) were used as received. Commercialized sulfuric acid prepared CNCs with a length of roughly 200 nm and a diameter of roughly 10 nm were purchased from ScienceK (CNC-S, Huzhou, China). The vegetable oil (Golden Dragon Oil, Wilmar International Limited Co.) used for preparation of emulsions contained 59% soybean oil, 21% rapeseed oil, 10% sunflower oil, 3% corn oil, 3% peanut oil, 3% rice oil, 0.6% sesame oil and 0.4% linseed oil. The deionized water purified by an AIKE Advanced-II-OS water purification system (Cheng Du Kangning Science and Technology Development Company) was used for all experiment.

Preparation of cellulose nanocrystals

The MCC was first wet with a certain amount of water and added to 75 wt% ZnCl_2 solutions containing certain amounts of HCl . The final concentration of ZnCl_2 was adjusted to 55, 60, or 65 wt% and the final HCl concentration was calculated to be 0.50, 0.75, or 1.0 M in the solution. The MCC solid to liquid ratio was fixed at 2.5 wt/v %. The MCC was hydrolyzed for 60 min at 80 °C under magnetic stirring. Unless otherwise specified, the reaction was terminated by pouring the mixtures into tenfold of icy water. The resultant cellulose gel was centrifuged at 12,000 rpm for 30 min at 5 °C (TGL-20000Cr, An Ting, Shanghai), washed with DI water three times, and then dialyzed using regenerated cellulose dialysis membranes (molecular weight cut off: 14, 000 Da) against DI water until

the conductivity of the solution outside the membrane reached a constant low value. Silver nitrate was added to the solution to confirm the absence of Cl^- (i.e. No white AgCl precipitate formed). The mixtures were then sonicated (40 Hz, 100 W, KH2200E, Kunshan Heli ultrasonic instruments Co., Ltd, China.) in an ice bath for 30 min, probe sonicated (300 W, SCIENTZ-IIID) for 5 min and then centrifuged at 3500 rpm for 15 min to separate fine CNC supernatant from large under hydrolyzed particles. Finally, the CNC suspensions were stored at 4 °C until further use. The CNCs were designated as CNC55-0.5, CNC55-0.75, CNC55-1.0, CNC60-0.5, etc., with the first number indicating w/w percent of ZnCl_2 , and the second number being HCl molar concentration.

Oil-in-water (O/W) Pickering emulsion

Each O/W Pickering emulsion was prepared by adding vegetable oil (0.9 mL) into 0.05, 0.1, 0.15, 0.25 or 0.3% CNC suspensions (2.10 mL) (oil: water=3:7). The mixture was sonicated (100 W) for 30 min and then let stand for 30 min. The top clear oil layer was carefully removed and its volume was subtracted from the initial oil volume (i.e. 0.9 mL) to obtain the emulsified volume in the suspension. The extent of emulsion was calculated from the ratio of emulsified volume to the initial oil volume. The droplet sizes in the Pickering emulsions are measured under an optical microscope (DM1000/DFC295). Each emulsion was diluted by adding 100 μL of emulsion to 10 mL of water and shaken by hand until uniformly dispersed, and one drop (10 μL) of the mixture was deposited on a glass slide for observation. The Nano Measurer software (Laboratory of Surface Chemistry and Catalysis, Department of Chemistry, Fudan University) was used to measure the droplet size. For each sample, the average size of the droplet was obtained from more than 50 individual droplets.

To evaluate the stability of the emulsions, O/W Pickering emulsions were prepared by adding vegetable oil (0.1 mL) into 0.1, 0.15 or 0.3% CNC suspensions (0.9 mL) (oil: water=1:9), and treated with the probe sonicator for 2 s (300 W). The samples were placed at 25 °C or 52 °C, and observed visually or under optical microscope during a 14-day period. Stability towards shear was evaluated by centrifugation of the freshly prepared emulsions (3500 rpm for

10 min or 5000 rpm for 10 min) (MC-12pro). The tubes were observed immediately after centrifugation.

Characterization

Scanning electron microscopy (SEM)

The morphology of the CNCs was studied using a scanning electron microscope (Hitachi, S-4800 Fe-Sem Japan). A small drop (10 μL , 0.05 wt%) of CNC suspension was deposited on aluminum foil, dried at 50 °C and sputter coated with a thin layer of platinum at 15 mA current (Hitachi E-1010 ion sputtering system, Japan). The sample was taped onto a SEM sample holder and imaged at 20–800,000 magnifications. The size of each type of CNC was measured and averaged from more than 100 particles using Image J software.

Energy-dispersive X-ray spectroscopy (EDS)

The elemental analysis of CNCs was performed using an EDS (HORIBA EX-250, Japan) attached to the SEM operating under 5000 magnification at 15 kV accelerating voltage and 15 mm working distance.

Atomic force microscope (AFM)

The morphology of the CNCs was also studied using an atomic force microscope (Bruker Dimension ICON, probe: RTESP-300). A drop of CNCs (10 μL , 0.05 wt%) was deposited onto the freshly cleaved mica surface, and dried under a far infrared lamp. The samples were scanned with the cantilever operating in the tapping mode, in the air, at resonance frequency of ~280 kHz.

X-ray powder diffraction (XRD)

The crystallinity and crystal structure of the MCC and freeze-dried CNCs were studied using an XRD powder diffractometer (D8 Advance, Bruker). Diffraction patterns were obtained with Cu $\text{K}\alpha$ radiation generated at 40 kV and 40 mA at a rate of 2°/min from 5° to 50° 2 θ . XRD spectra were analyzed

by MDI Jade 6.0 software. A cubic spline baseline reflecting the blank run was used to eliminate any background signal. Each crystalline and amorphous peak in the spectrum was fit to the Pearson VII model to obtain crystallinity index (CrI), crystal size, and the contents of cellulose I and cellulose II as reported previously (Xing et al. 2018, 2020).

Fourier transform infrared spectroscopy (FTIR)

A FTIR spectrometer (PerkinElmer, USA) was used to obtain the IR spectra of MCC and CNCs. The freeze-dried sample of 2 mg was grinded with 200 mg of KBr and pressed into a transparent pellet. Each spectrum was recorded at a resolution of 4 cm^{-1} over a range of $4000\text{--}400\text{ cm}^{-1}$ and averaged from 60 scans in transmittance mode.

X-ray photoelectron spectroscopy (XPS)

The elemental composition of the CNC surface was analyzed by XPS (Thermo Fisher Scientific K-Alpha). The radiation source was a monochromatic Al K α at an operation voltage of 15 kV and a current of 10 mA. The test energy was 1486.8 eV. The spot size was 30–50 μm . The sample chamber pressure during analysis was about 2×10^{-9} mbar.

Zeta potential measurement

The stability of the CNC suspensions was evaluated by a zetasizer (Nano ZSE Zetasizer, Malvern Instruments, Ltd). The zeta potential values (ζ , mV) of the aqueous suspensions (0.05 wt%) were determined in triplicates at 25 °C with a 780 nm wavelength laser.

Thermogravimetric analysis (TGA)

The thermal properties of the CNCs were analyzed with a thermogravimetric analyzer (TG209F1Li-braTM). Freeze-dried CNCs of 8–12 mg were weighed and placed into an Al_2O_3 crucible ($\varnothing 6.8 \times 4.0\text{ mm}$). Each sample was heated from 35 to 500 °C at a heating rate of 10 °C/min under nitrogen atmosphere (30 mL/min).

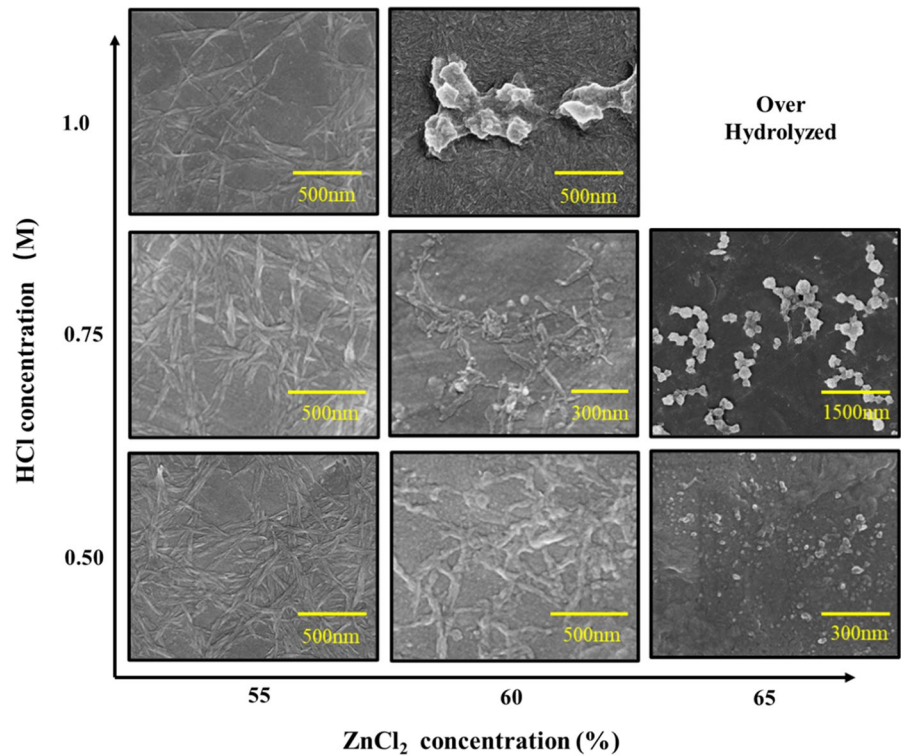
Results and discussion

Isolation of cellulose nanocrystals

Microcrystalline cellulose (MCC) was subjected to acidified zinc chloride treatments with ZnCl_2 and HCl concentration varying in ranges between 55–65% and 0.5–1 M, respectively. After that, low-intensity and short ultrasonic treatments (100 W for 30 min followed by 300 W for 5 min) were used to enhance the defibrillation level and dispersibility of the CNCs (Gu et al. 2017; Tang et al. 2014; Xing et al. 2018). Our preliminary results and previous studies all suggest that visually complete dissolution of MCC occurs when the ZnCl_2 concentration is higher than 65% (i.e. $\text{ZnCl}_2 \cdot 4\text{ H}_2\text{O}$, without adding acid) under a temperature of 80 °C (Bodachivskyi et al. 2020; Leipner et al. 2000). Thus ZnCl_2 of 55%, 60%, and 65% concentrations selected in this study correspond to swelling or dissolving cellulose to varied degrees. It was supposed that if acid was introduced at an accurate low amount to preferably react with less ordered regions in cellulose, CNCs with different allomorphs and/or morphology would be obtained.

Cellulose remained in a solid-state in 55% ZnCl_2 at any acid concentration and the large MCC particles were gradually hydrolyzed to nano-sized CNCs. The reaction mixtures remained ivory white or light yellow which indicated the cellulose was not dissolved. When ZnCl_2 concentration was increased to 60%, the reaction mixtures first appeared ivory white, and then became transparent, indicating swelling and dissolution of cellulose chains occurred. The transparent mixtures turned into ivory white again after being poured into tenfold icy water, suggesting a regeneration process occurred. If the ZnCl_2 concentration was 65%, the reaction mixtures quickly became transparent due to the dissolution of MCC. Regenerated CNCs were produced when the HCl concentration was 0.5–0.75 M. As the HCl concentration was raised to 1 M, no visible solid residue was regenerated even after the centrifugation, indicating the cellulose was over hydrolyzed to small oligomers or monosugars.

The morphology of the obtained CNCs was analyzed using SEM imaging (Fig. 1). The dimensions and size distributions of the CNCs are recorded in Table 1 and Fig. S2. At low ZnCl_2 concentration

Fig. 1 SEM images of CNCs**Table 1** Morphology, dimensions and aspect ratio of CNCs

Samples	Morphology	Length ^a (nm)	Width ^a (nm)	Diameter ^a (nm)	Aspect ratio
CNC55-0.5	Rod-like	279.3 ± 110.0	26.8 ± 17.2	–	10.4
CNC55-0.75	Rod-like	181.3 ± 110.1	18.0 ± 11.0	–	10.1
CNC55-1.0	Rod-like	233.2 ± 130.1	20.7 ± 14.0	–	11.3
CNC60-0.5	Rod-like (majority) and sphere-like	180.7 ± 80.2	28.5 ± 20.2	156.3 ± 90.5	6.3
CNC60-0.75	Rod-like (majority) and sphere-like	139.8 ± 80.7	19.7 ± 15.4	63.2 ± 31.8	7.1
CNC60-1.0	Rod-like and sphere-like	148.2 ± 80.2	17.0 ± 11.3	253.8 ± 103.3	8.7
CNC65-0.5	Sphere-like	–	–	25.8 ± 12.8	–
CNC65-0.75	Sphere-like with two size populations	–	–	292.3 ± 150.2/85.2 ± 44.8	–

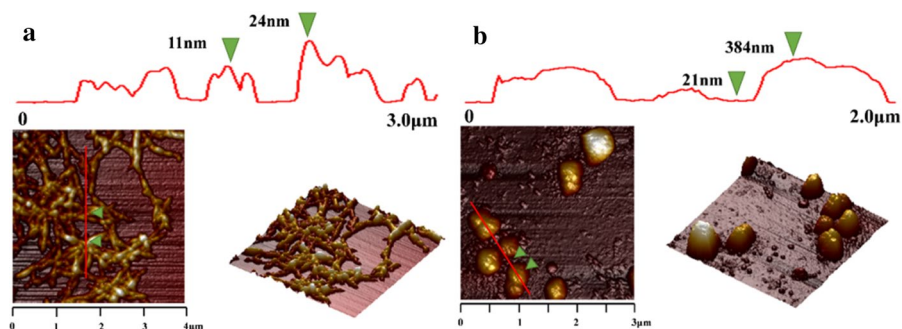
^aDimensions were measured from the SEM images

(55%), rod-like or spindle shaped CNCs could be found. The length and width of CNC55-0.5 ranged from 160 to 480 nm and 12–42 nm, averaged 279.3 ± 110.0 nm and 26.8 ± 17.2 nm, respectively. With increased HCl concentration, CNC55-0.75 and CNC55-1.0 exhibited relatively smaller dimensions than that of CNC55-0.5. CNC55-0.75 had a length of 181.3 ± 110.1 nm and a width of 18.0 ± 11.0 nm while CNC55-1.0 exhibited a length of 233.2 ± 130.1 nm and a width of 20.7 ± 14.0 nm. The size of the

rod-like CNCs prepared using 55% zinc chloride was comparable with CNCs prepared with hydrochloric acid or a mixed acid containing hydrochloric acid (length: 200–300 nm, width: 10–30 nm) (Gu et al. 2013; Braun et al. 2008; Yu et al. 2016).

When ZnCl_2 concentration was increased to 60%, although rod-like CNCs were still dominant, particles of spherical and granular shape appeared (also see Fig. S3). The lengths of the rod-like CNC60-0.5, CNC60-0.75, and CNC60-1.0 were 180.7 ± 80.2 nm,

Fig. 2 AFM topography images of representative rod-like and sphere-like CNCs: **a** CNC55-0.5, **b** CNC65-0.75



139.8 ± 80.7 nm, and 148.2 ± 80.2 nm, respectively, while their widths were 28.5 ± 20.2 nm, 19.7 ± 15.4 nm, and 17.0 ± 11.3 nm, respectively. In CNC60-1.0, large irregular and sphere-like particles with an average diameter of 253.8 ± 103.3 nm were frequently observed in SEM images. A small portion of spherical CNCs could also be found in CNC60-0.5 and CNC60-0.75 samples (Fig. S3a-b), but their size was much smaller (CNC60-0.5: 156.3 ± 90.5 nm, CNC60-0.75: 63.2 ± 31.8 nm). Compared with CNCs produced at 55% ZnCl_2 , rod-like CNCs generated at 60% ZnCl_2 generally exhibited smaller sizes and lower aspect ratio. This result is consistent with the increasing acidity and cellulose dissolubility in the zinc chloride hydrate with reducing levels of water present (Bodachivskyi et al. 2020). High acidity and acid penetration level accelerate the hydrolysis of cellulose.

At high ZnCl_2 concentration (65%), only sphere-like nanoparticles were observed when 0.5–0.75 M HCl was added. When the HCl concentration was 1.0 M, the cellulose was over hydrolyzed and no solid particles could be regenerated. The diameter of CNC65-0.5 was in the range of 12–36 nm with an average value of 25.8 ± 12.8 nm. The size of CNC65-0.75 (150–420 nm, mean: 292.4 ± 150.2 nm) was much larger than that of CNC65-0.5 and their morphology was similar to those sphere-like CNCs found in CNC60-1.0. Interestingly, a few smaller granular and sphere-like nanoparticles with a diameter of 35–130 nm were also observed in CNC65-0.75 (Fig. S3d). In previous studies, spherical CNCs with a size larger than 200 nm were seldom reported in the literature (Liu et al. 2021; Tian et al. 2022). A few studies suggest that the cellulose nanospheres form via the self-assembly of shorter/smaller cellulose nanoparticles through interfacial hydrogen bonds

(Beaumont et al. 2019; Chen et al. 2019; Lu and Hsieh 2010; Verma et al. 2021). To test this hypothesis, MCC hydrolyzed by 65% ZnCl_2 and 0.75 M HCl was poured into 10 times 0.05% Tween 20 solution instead of DI water. Other treatment steps were kept the same as for preparation of CNC65-0.75. Tween 20 is a polysorbate-type nonionic surfactant and is able to prevent non-specific aggregations of nanoparticles. If aggregation of smaller cellulose nanoparticles occurs during the water regeneration step, Tween 20 may disrupt this self-assembly process. However, as shown in Fig. S3e, large sphere-like nanoparticles with a diameter of 200–900 nm were still produced, indicating they were not formed from self-assembly of the small nanospheres. Their formation mechanism was discussed later in this study.

The AFM images of representative rod-like and sphere-like CNCs were shown in Fig. 2. The CNC55-0.5 particles displayed the typical spindle-like structure with thinner ends, which were similar to CNCs produced by hydrochloric acid hydrolysis of MCC (Guo and Catchmark 2012). The thickness of the CNC55-0.5 was in a range of 5–24 nm, generally close to the width as measured by SEM (Fig. 1) and consistent with typical rod-like CNC (Ingvald Kvien et al. 2005). The CNC65-0.75 particles exhibited sphere-like structure in two size populations: much larger 150–500 nm diameter spheres and smaller particles with a diameter of less than 100 nm. The height of the larger spheres was in a range of 119–565 nm, very close to the diameter determined by SEM (Fig. 1). This result indicates the sphere-like CNCs produced in this study might be different from the disk-like CNCs reported in previous studies (Torlopov et al. 2018; Yang et al. 2013). Most spherical CNCs reported in the literature were not characterized by AFM imaging (Tian et al. 2022), which is

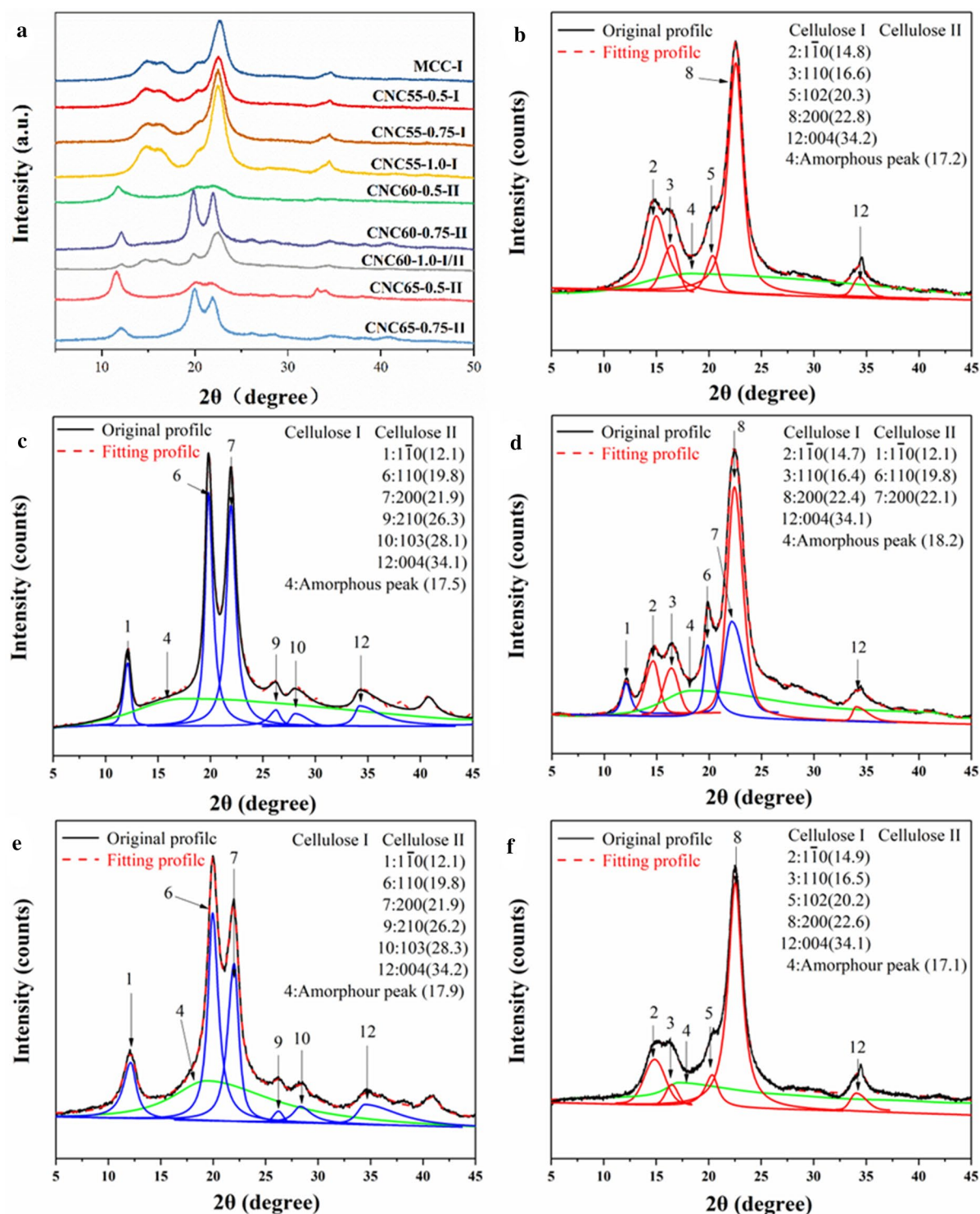


Fig. 3 a XRD profiles of MCC and CNCs prepared under various conditions, **b–e** X-ray diffractogram analysis of MCC and representative CNCs: **b** CNC55-0.5, **c** CNC60-0.75, **d** CNC60-1.0, **e** CNC65-0.75, and **f** MCC

Table 2 Crystal allomorph, crystallinity, thermal property and zeta potentials of MCC and CNCs

Samples	Cellulose allomorph	Cr I ^a (%)	Moisture ^b (%)	Char ^b (%)	T _n ^b (°C)	T _{max} ^b (°C)	ζ potential ^c (mV)
MCC	I	79.9	1.9	10.0	217.4	336.2	–
CNC55-0.5	I	64.2	1.9	9.1	180.4	334.8	– 16.5
CNC55-0.75	I	62.1	1.9	8.2	203.1	336.3	– 18.5
CNC55-1.0	I	68.3	1.5	10.9	201.2	343.5	– 18.1
CNC60-0.5	II	49.6	2.8	21.3	170.2	325.1	– 14.7
CNC60-0.75	II	55.6	3.4	19.5	180.5	316.3	– 15.4
CNC60-1.0	I/II	67.9	2.4	23.3	188.9	341.1	– 16.3
CNC65-0.5	II	56.1	3.1	32.9	202.3	319.1	– 12.0
CNC65-0.75	II	68.4	3.4	35.7	187.6	330.7	– 20.5

^aCrystallinity was calculated from the XRD profiles^bMoisture content, char residue, T_n and T_{max} were calculated from the TGA curves^cZeta potential was obtained using the dynamic light scattering particle size analyzer

commonly used to probe the height of the nanoparticles. Thus it is debatable whether the derived “spherical CNC” possesses a micro morphology of “sphere” or “disc” (Jiang et al. 2021). The spherical CNC65-0.75 nanoparticles confirmed by AFM height profile may be associated with a relatively rigid structure.

Crystalline structure

The crystal structure of the MCC and the CNCs was analyzed by powder XRD (Fig. 3 and Fig. S4). Depending on the reaction conditions, typical cellulose I, cellulose II or mixture of cellulose I and II profiles were observed. With 55% ZnCl₂, CNCs always exhibited typical cellulose I XRD patterns as that of MCC, with the characteristic diffraction peaks at 2θ angles of about 15.0, 16.5, and 22.5° corresponding to the cellulose I (110), (110), and (200) crystal planes, respectively (French 2014). When ZnCl₂ concentration was increased to 60–65%, the resultant CNCs showed diffraction peaks at 12.2, 20.0, and 22.1°, which were related to cellulose II (110) (110) and (020) lattice planes, respectively (French 2014). Interestingly, CNC60-1.0 was identified as a mixture of cellulose I and II. It is well known that the dissolution of cellulose in ZnCl₂ molten salt hydrates is accompanied by a cellulose allomorph transformation from cellulose I to cellulose II. However, swelling of the cellulose can also result in a change of the allomorph depending on the water content and the acidity (Leipner et al. 2000). ZnCl₂ of 55% represented

a weak swelling agent and thus the resultant CNCs remained cellulose I. A stronger swelling can be stated for the 60% ZnCl₂ and the regenerated CNCs were cellulose II or mixed cellulose I/II. Further increase of the ZnCl₂ content to 65% led to a dissolution of cellulose and the regenerated CNCs were 100% cellulose II.

According to our previous study (Xing et al. 2018), a peak deconvolution method was used to estimate the contents of cellulose I and cellulose II components and the crystallinity of the samples (Table 2, Fig. 3, and Fig. S4). The cellulose I content in CNC60-1.0 was 71.6%, while all the other CNCs were pure cellulose I or II. The crystallinity (CrI) of all the CNCs (49.6–68.4%) were lower than that of the MCC (CrI: 79.9%). At fixed ZnCl₂ concentration, increasing the HCl concentration generally contributed to higher CrI, indicating that the less ordered cellulose was hydrolyzed relatively fast at higher acid concentration. Compared to the nanocellulose prepared via mechanochemical synergy method using ZnCl₂·3H₂O (CrI: 33.5%) (Lu et al. 2019), the CNCs obtained in our study exhibited much higher crystallinity. Low ZnCl₂ concentration and the addition of a small amount of acid contributed to high degree of order in CNCs.

The average crystalline dimensions perpendicular to the (110), (110) and (200)/ (020) planes of CNCs were estimated using the Scherrer's equation (Table S1). Only samples with high crystallinity and single allomorph were selected for the calculation.

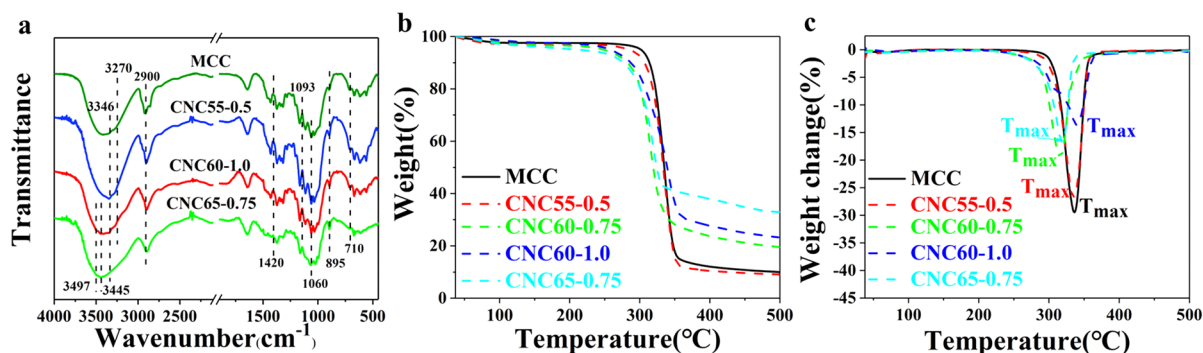


Fig. 4 **a** FTIR spectra, **b** TGA, and **c** DTGA curves of MCC and CNCs

The crystal size (D_{200}) of all cellulose I nanocrystals (4.6–5.0 nm) did not change significantly compared with MCC (4.7 nm). Larger crystallites were measured in the case of the cellulose II nanocrystal, values of 4.9–7.8 nm being found along any of the ($\bar{1}10$), (110) and (020) planes of CNC65-0.75. This result indicates with increased HCl acid concentration, CNC65-0.75 formed well-organized crystallites.

Chemical and thermal properties of CNCs

The crystalline allomorph of CNCs was also verified by FTIR. The spectra of representative CNC I (CNC55-0.5), CNC I/II (CNC60-1.0), CNC II (CNC65-0.75) and MCC were compared and shown in Fig. 4a. MCC and all CNCs exhibited typical cellulose peaks at 3346, 2900, and 1060 cm^{-1} , representing stretching vibration of O–H, C–H, and C–O bonds, respectively. The spectra of CNC60-1.0 and CNC65-0.75 displayed two additional weak peaks at 3497 and 3445 cm^{-1} relating to the intramolecular hydrogen bonds of cellulose II (Han et al. 2013). The absorption bands at 3270, 1093 and 710 cm^{-1} , which were characteristic of O–H stretching, planar stretching vibration of ring skeleton, and O–H out-of-plane bending of cellulose I β (Yamamoto et al. 1996), were found in MCC, CNC55-0.5, and CNC60-1.0, but disappeared in CNC65-1.0. Compared to MCC and CNC55-0.5, the symmetric bending vibration peak of CH_2 at 1432 cm^{-1} in CNC60-1.0 and CNC65-0.75 shifted to a lower wave number at 1420 cm^{-1} , which suggested the formation of new hydrogen bonds and change of CH_2OH conformation in cellulose II structure (Li et al. 2015). The band at 895 cm^{-1} assigned as asymmetric stretching vibration peak of C–O–C

increased in intensity in CNC60-1.0 and CNC65-0.75, corresponding to the appearance of cellulose II. In summary, FTIR spectroscopy confirmed CNC55-0.5 was cellulose I as MCC. CNC60-1.0 was cellulose I and II hybrid while CNC65-0.75 was cellulose II. The IR spectra of additional CNC samples were shown in Fig. S5. Analysis of IR characteristic bands showed consistency with the crystalline allomorph results determined by XRD.

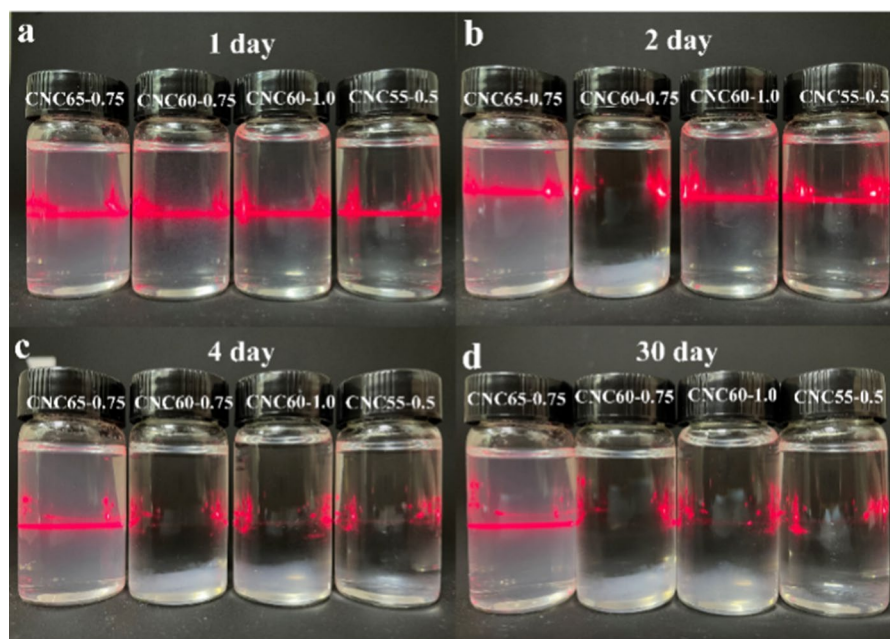
EDS spectra were acquired from CNC60-1.0, CNC65-0.5 and CNC65-0.75 (Fig. S6). Carbon, oxygen, aluminum and platinum were found in all three samples. While the carbon and oxygen were from cellulose, the aluminum and platinum came from the SEM sample mount and coating, respectively. A small amount of zinc was detected in CNC65-0.5 (2.1 at%) and CNC65-0.75 (2.0 at%), but not in CNC60-1.0. Simultaneously, chlorine was found in CNC65-0.75 (4.2 at%), almost double to the Zn element in the same sample. However, chlorine was neither detected in CNC65-0.5 nor CNC60-1.0. As reported in Experimental Section, all CNC samples were dialyzed again DI water for an extended period and AgNO_3 was added to confirm the absence of Cl^- in solution. Since no Zn was detected in CNC60-1.0, it is safe to state that the Zn^{2+} on the surface of nanoparticles could be removed. While 65% ZnCl_2 solution was used, hydrated Zn^{2+} penetrated deeply into the cellulose crystallites and coordinated with the cellulose hydroxyls. During the regeneration process, a small amount of Zn^{2+} may retain inside the CNCs. The zinc chloride may covert to zinc hydroxide or tetrabasic zinc chloride in the process. The remaining of zinc in cellulose even after several thorough washes was also observed in a previous report (Sen et al. 2013).

Elements present in CNCs were also confirmed by XPS. Survey spectra were acquired from CNC60-1.0, CNC65-0.5, and CNC65-0.75 (Fig. S7a). The peaks assigned for C 1s and O 1s elements were present in all three samples. Zn and Cl elements were not found in CNC60-1.0 and CNC65-0.5, but detected in CNC65-0.75. The C 1s spectrum was fit to three contributions in all three CNCs (Fig. S7b): (1) C linked to C or H, namely C–(C, H), with electron binding energy of 284.8 eV. Since cellulose does not have any aliphatic carbon, this feature is usually associated with impurities present on the surface of the CNCs; (2) C linked to O, namely C–O, with electron binding energy of 286.6 eV; (3) C connecting two O, namely O–C–O or C=O, with electron binding energy of 288.0 eV. The C 1s spectra of all three CNC samples looked similar, indicating the presence of Zn^{2+} in CNC65-0.75 did not significantly change the chemical state of carbon in cellulose. The high resolution O 1s spectra are shown in Fig S7c. The O 1s peak in CNC60-1.0 and CNC65-0.5 was mainly attributed to C–O and O–H with binding energy of 533.0 eV. In CNC65-0.75, the O 1s peak became wider and shifted to lower binding energy, indicating oxygen had more chemical states in this sample. The O 1s spectrum was thus fit to three contributions: (1) a low-binding-energy component at 531.8 eV, corresponding to Zn–O in the metal oxide; (2) C–O at 532.7 eV; (3) a high-binding-energy component at 533.3 eV, attributed to the hydroxyl contents from Zn–OH. The two strong XPS peaks exhibited at 1045.1 and 1022.1 eV in CNC65-0.75 were ascribed to the binding energy of Zn $2p_{1/2}$ and Zn $2p_{3/2}$, respectively, confirming the presence of Zn^{2+} (Fig. S7d). The Zn $2p_{3/2}$ peak at 1022.1 eV could be assigned to Zn^{2+} species in ZnO (1021.8 eV) (Battistoni et al. 1981), ZnCl_2 (1021.9 eV) (Klein and Hercules 1983) or $\text{Zn}(\text{OH})_2$ (1021.8 eV) (Deroubaix and Marcus 1992). The observed spin–orbit splitting of Zn 2p (between Zn $2p_{3/2}$ and $2p_{1/2}$) was approximately 23.0 eV, consistent with the value of ZnO. The Cl $2p_{3/2}$ (198.6 eV) and Cl $2p_{1/2}$ (200.8 eV) (Fig. S7e) belonged to a metal chloride, which confirmed that Zn and Cl existed in CNC65-0.75 as ZnCl_2 (Bankar et al. 2019). However, the content of Zn (5.1 at%) was higher than that of Cl (0.4 at%), indicating that Zn might also exist as zinc oxide, zinc hydroxide or tetrabasic zinc chloride. The XPS results are consistent with the EDS analyses. With 65% ZnCl_2 and especially high

acid concentration, Zn^{2+} retained in CNCs, possibly by physical entrapment. Both XPS (several nm) and EDS (hundreds of nm) detect the surface composition of the CNCs. Since both techniques are semi-quantitative, the exact atomic percentage of Zn and Cl elements and their distribution are difficult to obtain.

The thermal stability of CNC samples with different cellulose allomorphs and shapes was evaluated using TGA (Fig. 4b, c, Table 2). TGA curves of all the samples showed small initial mass loss around 100 °C resulting from evaporation of adsorbed moisture, followed by the most significant mass loss due to decomposition of organic molecules and final mass loss from charring (Achoundong et al. 2013; Fortunati et al. 2014). The moisture contents, as measured by the weight loss at 100 °C, were 1.9, 1.9, 3.4, 2.4 and 3.4% for MCC, CNC55-0.5, CNC60-0.75, CNC60-1.0 and CNC65-0.75, respectively. CNCs with more cellulose II content contained more moisture. Degradation of MCC initially occurred (T_n) at about 217.4 °C and achieved its peak (T_{max}) at around 336.2 °C, both higher than those of most CNCs. The slightly lower thermal stability of CNCs was ascribed to the reduced molecular weight and crystallinity. At the same ZnCl_2 concentration, CNCs with high crystallinity (e.g. CNC55-1.0, T_{max} : 343.5 °C and CNC60-1.0, T_{max} : 341.1 °C) exhibited higher thermal stability than those CNCs with low crystallinity (e.g. CNC55-0.5, T_{max} : 334.8 °C and CNC60-0.5, T_{max} : 325.1 °C). Although cellulose II is a thermodynamically more favorable allomorph, the CNC II nanoparticles (e.g. CNC60-0.75, T_{max} : 316.3 °C and CNC65-0.75, T_{max} : 330.7 °C) were less thermally stable than the CNC I nanoparticles (e.g. CNC55-0.75, T_{max} : 336.3 °C) perhaps because the CNC II nanoparticles had smaller sizes, especially those containing tiny nanospheres. Interestingly, the DTG curve of CNC60-1.0 showed a main peak at 341.1 °C and a small shoulder at 317.0 °C, consistent with its composition of 71.6% cellulose I and 28.4% cellulose II. The thermal stability of acidified ZnCl_2 -prepared CNCs (T_{max} : 316.3–343.5 °C) were much higher than that of H_2SO_4 -prepared CNCs (T_{max} : ~255 °C) (Morán et al. 2007) and comparable with that of HCl-prepared CNCs (Cudjoe et al. 2017; Zhang et al. 2014). The amount of char residue (at 500 °C) of CNC I nanoparticles (8.2–10.9%) was close to that of MCC (10.1%). The char residue of CNCs prepared with 60% ZnCl_2 increased to 19.5–23.3%, and

Fig. 5 Suspension stability of CNCs after settling for 1, 2, 4 and 30 days. From left to right are CNC65-0.75, CNC60-0.75, CNC60-1.0, and CNC55-0.5



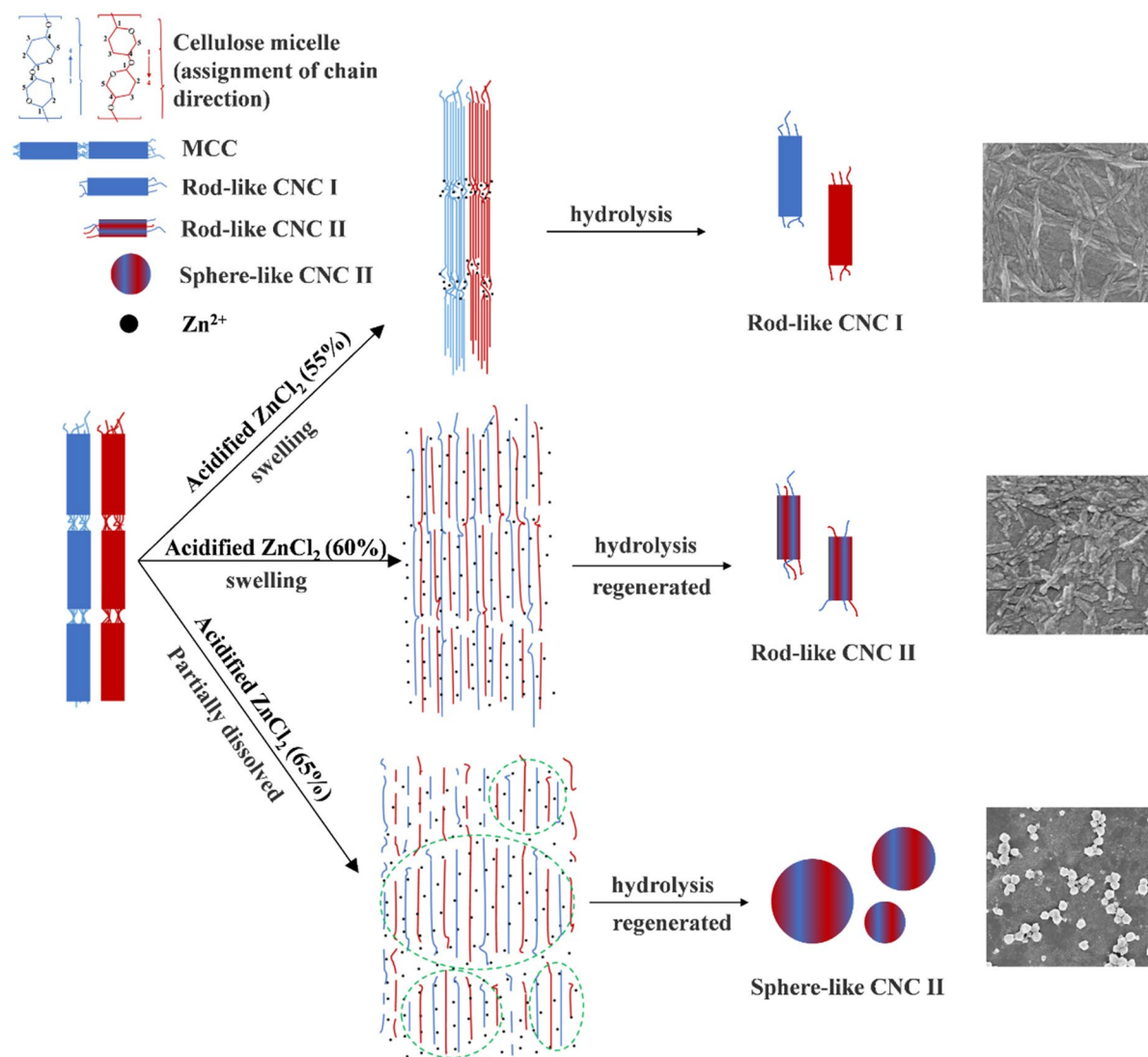
the value further increased to 32.9–35.7% for CNC II obtained using 65% ZnCl_2 . The char residue of CNC II was generally more than that of CNC I. CNC II with small size and more free-end chains degraded at lower temperature, and thus facilitated the char residue formation (Liu et al. 2021; Lu et al. 2019; Wang et al. 2007; Xing et al. 2018). The entrapment of zinc in CNCs prepared at high ZnCl_2 concentration also led to increased char content.

Suspension stability

The stability of CNC suspensions is critical for their applications in nanocomposites and emulsion stabilizers. Figure 5 shows the photographs of representative CNC samples (0.3 wt%): CNC55-0.5, CNC60-0.75, CNC60-1.0 and CNC65-0.75 after sonication for 1 day, 2 days, 4 days and 30 days. All the CNC suspensions remained stable in 1 day. The Tyndall effect was observed when a beam of red laser (wavelength: 650 nm) passing through the suspensions (Fig. 5a). This phenomenon results from the light scattering when the dispersed particle size is below or near the wavelength of the incident light. After 2 days, sedimentations were observed in CNC60-0.75 (Fig. 5b). Most CNC55-0.5 and CNC60-1.0 precipitated in 4 days (Fig. 5c). The suspension of CNC65-0.75 gave the best stability without any sedimentations for more

than 30 days. The sample still presented the Tyndall effect after a long stand, indicating that the CNC65-0.75 dispersed stably in aqueous. Since the obtained CNCs were separated from large under hydrolyzed particles by centrifugation during the preparation process, they were relatively stable against gravity force. Even for the precipitated CNCs, good suspensions can be again achieved through ultrasonic treatment.

The zeta potential of CNC suspensions (0.05 wt%), which was used to estimate the effect of charge groups on CNC suspension stability, was shown in Table 2. The electrostatic repulsion between nanoparticles is believed to stable most aqueous colloidal suspensions. Larger absolute values of zeta potential are associated to better suspension stability. The zeta potentials of the CNCs suspensions fell into the range of -20.5 to -12.0 mV. Generally, at the same ZnCl_2 concentration, increasing the HCl concentration often led to more negative zeta-potential. The zeta potential of CNC55-0.5, CNC60-0.75, CNC60-1.0 and CNC65-0.75 were -16.5 , -15.4 , -16.3 and -20.5 mV, respectively, well reflecting their suspensions stability (Fig. 5). The HCl-prepared CNCs reported previously usually have zeta potentials of -16 to -12 mV, larger than those of the H_2SO_4 -prepared CNCs (< -20 mV) (Yu et al. 2013). HCl treatment did not result in any cellulose chemical modification. H_2SO_4 treatment introduced



Scheme 1 Proposed formation mechanism of CNC I and CNC II nanoparticles in acidified zinc chloride

sulfate groups onto the cellulose surface (Gu et al. 2013), which leads to more stable aqueous suspensions (Araki et al. 1998). CNCs prepared by acidified ZnCl_2 treatment of MCC exhibited comparable zeta potentials to HCl-prepared CNCs. ZnCl_2 molten salt is a non-derivatizing cellulose solvent, and thus it is rational that the CNCs obtained had little chemical modification and exhibited relatively high zeta potentials. The relatively low zeta-potential of CNC65-0.75 perhaps due to the inclusion of zinc chloride and as a result, CNC65-0.75 displayed the best suspension stability.

Proposed formation mechanism of CNC I and CNC II

The proposed formation mechanism of CNC I and CNC II with various shapes and sizes was given in Scheme 1. In our tentative model, the concentration of ZnCl_2 plays a significant role in controlling the swelling levels of the MCC and consequently, the hydrolysis of cellulose chains. When low ZnCl_2 concentration (55%) was used, the hydronium and zinc ions penetrated slowly from the surface to the inner disordered regions of cellulose. Zinc ions had some chance to coordinate with the hydroxyls of cellulose

and facilitated the disruption of cellulose intermolecular hydrogen bonds. Hydronium protonated the oxygen atom in the β -1, 4 glycosidic bonds and led to their destruction. Cellulose I crystalline cores still retained and finally rod-like CNC I formed. With increased HCl concentration, highly crystalline CNCs were produced by the relatively faster degradation of disordered regions than crystalline regions. It was an acid-catalyzed heterogeneous hydrolysis.

Increasing ZnCl_2 concentration to 60% resulted in cellulose allomorph transformation under a swelling condition. This process may be similar as the mercerization-induced allomorph transformation of cellulose (Li et al. 2021) and different from cellulose dissolution with all the cellulose chains disintegrated. Hydrated Zn^{2+} and H^+ ions penetrated into the cellulose crystallites. Zn^{2+} -cellulose complex formed and disintegrated the cellulose microfibrils into layers of mobile crystalline planes, which were composed of parallel cellulose chains assembled via inter-chain hydrophobic interactions (Lindman et al. 2017). Exchanging of the crystalline planes induced an anti-parallel conformation of cellulose chains cross the planes. Swelling of cellulose chains promoted the hydrolysis of the β -1, 4 glycosidic bonds in the newly formed disordered regions. The resultant rod-like CNCs were relatively shorter than the CNCs produced under low ZnCl_2 concentration. A very few of sphere-like CNCs were also produced under this condition. The increased accessibility of cellulose caused mild hydrolysis occurred at both the crystalline and disordered regions and contributed to the formation of sphere-like particles (Wang et al. 2008; Yang et al. 2013). Surprisingly, while CNC60-0.5 and CNC60-0.75 were cellulose II, CNC60-1.0 was a mixture of cellulose I and II. Perhaps at relatively high acid concentration, a portion of swollen and disordered cellulose hydrolyzed fast into small molecules and could not recrystallize to form cellulose II, but a portion of cellulose I crystalline cores still remained. Another possibility is that the excess Cl^- disrupted the formation of hydrated Zn^{2+} and thus the Zn^{2+} -cellulose complex.

When the concentration of ZnCl_2 was raised to 65%, dissolution of cellulose occurred due to the coordination of hydrated Zn^{2+} with cellulose hydroxyls. Under mild hydrolysis conditions (0.5–0.75 M HCl), cellulose was hydrolyzed into sphere-like CNCs with varied sizes as a result of increasing

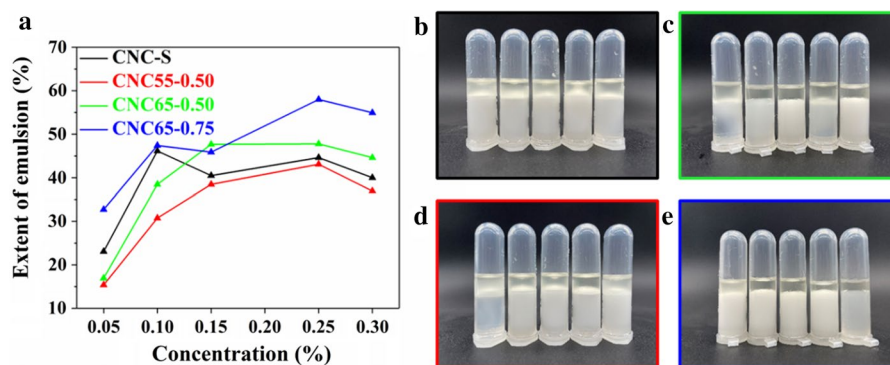
accessibility of MCC. As discussed above, large sphere-like CNCs produced in this study were not from self-assembly of small nanoparticles. During high concentration ZnCl_2 treatment, the swollen and dissolved cellulose might form random sub-micrometer size block structure (Liu et al. 2021). The MCC dissolved in 65% ZnCl_2 still remained some degree of order under the reaction time scale. Exchange of crystalline planes or entanglement of cellulose chains might occur inside a block. Hydrolysis of cellulose transformed this blocky structure into sphere-like particles by trimming the edge of the ordered structure. When washed with water, most Zn^{2+} and Cl^- ions were removed, while a few still remained in the newly regenerated cellulose II crystallite. Additional studies are deserved to understand the formation of spheres and their size evolution mechanism.

Therefore, in the acidified ZnCl_2 system, the accessibility of cellulose and the influence of hydrolysis were two determining factors in the particle size and morphology of the obtained CNCs. It is demonstrated that rod-like CNC I and CNC II as well as sphere-like CNC II could be easily produced by tuning the ZnCl_2 and HCl concentration. The concentration of acid used (0.5–1 M) was relatively low compared to the classical mineral acid hydrolysis methods (e.g. 4–6 M HCl) (Araki et al. 1998; Yu et al. 2013). It is worthy to note that the ZnCl_2 is recyclable and reusable after the hydrolysis (Lu et al. 2019), even though the recovery of reagents is not the main focus of this work. As far as we are aware, this is the first study showing highly crystalline CNCs with tunable shapes and allomorphs could be obtained via facile acidified ZnCl_2 treatment.

CNC-stabilized oil-in-water emulsions

The spherical CNCs could be beneficial in applications such as Pickering emulsion stabilizers (Dong et al. 2021; Liu et al. 2020), bioassay probes (Choi et al. 2015; Zhang et al. 2018), and drug/catalysis carriers (Carrick et al. 2014). In this work, emulsions stabilized by different CNCs were compared. Spherical CNC65-0.5 and CNC65-0.75 as well as rod-shaped CNC55-0.5 and commercialized CNC-S were added into the vegetable oil and water mixtures to generate Pickering emulsions. Being globally hydrophilic materials, unmodified CNCs only stabilize o/w emulsions. Biphasic solution with a top oil layer

Fig. 6 **a** Extent of Pickering emulsion, **b–d** photographs of Pickering emulsion: CNC-S (**b**), CNC55-0.5 (**c**), CNC65-0.5 (**d**) and CNC65-0.75 (**e**). The concentrations of CNCs are 0.05, 0.10, 0.15, 0.25 and 0.30% from left to right



formed in 3/7 v/v oil/water. After sonication (100 w, 30 min), the transparent aqueous layer turned into creamy opaque O/W emulsion, while a thin layer of oil still remained on the top (Fig. 6). With increasing CNC concentration from 0.05 to 0.3%, the extent of emulsion in all samples first increased and then decreased slightly. Rod-like CNC-S and CNC55-0.5 showed very similar trend. The extent of emulsion increased from 15 to 43% with increasing CNC55-0.5 concentrations from 0.05 to 0.25%, then decreased to 37% at 0.3% concentration. For CNC-S, the extent of emulsion increased from 23 to 46% when the concentrations increased from 0.05 to 0.1%, then decreased to roughly 40% at higher CNC concentration. The sphere-like CNCs generally stabilized larger volume of emulsion. The extent of emulsion increased from 33 to 58% with increasing CNC65-0.75 concentrations from 0.05 to 0.25%, then remained high at 56% up to 0.3% concentration. CNC65-0.5 exhibited lower extent of emulsion than CNC65-0.75, but still showed 48% extend of emulsion at 0.15% concentration. The light microscope images of emulsion droplets are shown in Fig. S8. The diameters of the droplets in CNC-S, CNC55-0.5, CNC65-0.5 and CNC65-0.75 stabilized emulsions were 2.74–3.97, 2.18–4.05, 1.48–4.61 and 1.64–3.28 μm , respectively. Generally, the size of the emulsion droplets gradually decreased as the concentration of CNCs increased. At high CNC concentration (0.25–0.30%), the emulsions stabilized by sphere-like CNCs exhibited a smaller droplet size than those with rod-like CNCs.

The emulsification method and energy input also play important role in the emulsion formation. Treated by probe sonication (300 W, 2 s), emulsions with 0.1%, 0.15% and 0.3% CNCs of different types formed in 1/9 v/v oil/water. The emulsion

ratio reached 100% and most emulsions were stable without phase separation for a few hours. With high energy ultrasound treatment, smaller emulsion droplets formed and cellulose nanoparticles dispersed better. As a result, more oil water interface was created and stabilized by CNCs. The emulsions were visually observed for 14 days (Fig. S9a). In all samples, two distinct phases formed within 2 days after emulsification. Thin creaming layers appeared on the top, and the lower layers were a turbid phase. After preparation for 2–14 days, for emulsions with rod-like CNC-S and CNC55-0.5, the creaming layer became thinner and the lower phase became less opaque with time. For emulsion with CNC65-0.5, coalescence occurred and a clear oil layer appeared on the top after 4 days. The appearance of the emulsions stabilized with CNC65-0.75 did not change further after 2 days. Micrographs taken after 14 days demonstrated that the creaming layers in emulsions with CNC-S, CNC55-0.5, and CNC65-0.75 contained a large number of oil droplets, while the turbid phase below the creaming layers were also emulsions contained well dispersed droplets (Fig S9c). More numerous droplets were found in CNC65-0.75 stabilized emulsions than in the CNC-S and CNC55-0.5 counterparts. However, the turbid phase in CNC65-0.5 only contained a small number of large oil droplets, confirming the coalescence of the emulsion. These results indicate emulsions prepared with CNC65-0.75 were remarkably stable against coalescence over time.

Emulsions were also tested for their stability towards heating to 52 ± 2 °C (Fig S9b). Interestingly, emulsions with CNC-S and CNC55-0.5 showed good stability for 6 days, but clear oil layers appeared on the surface afterwards. The remaining lower emulsion phase still contained well dispersed small oil droplets

after 14 days (Fig S9d). In CNC65-0.5 and CNC65-0.75, two distinct phases formed within 2 days after emulsification. The top layer was a creaming layer, and the lower layer was a turbid phase (Fig. S9b). The emulsion with CNC65-0.5 showed evidence of coalescence after 6 days. The creaming phase disappeared and the mixture became less opaque. The optical microscope image confirmed that the turbid phase contained only coalescent large oil droplets (Fig. S9d). No visual and microscopic signs of change were seen in the CNC65-0.75 stabilized emulsion after initial creaming. These results indicate emulsions prepared with CNC65-0.75 were also stable toward heating.

The centrifuge stability of the freshly prepared emulsions was evaluated and shown in Fig. S10. No appearance changes were found in all emulsions when the centrifugal speed was 3500 rpm, indicating these emulsions had excellent stability toward shear. With increasing centrifugal speed to 5000 rpm, a small amount of precipitation appeared in most samples except the one with 0.3% CNC65-0.75. These precipitates were resulted from flocculation of emulsion droplets and they gradually floated to the top of the tubes. In most samples, however, coalescence occurred and an oil fraction as top layer was visible.

The surface chemistry, shape and size of nanocellulose all have influence on emulsion formation (Fujisawa et al. 2017; Madivala et al. 2009). Both CNC-S and CNC55-0.5 displayed a rod-like morphology with lengths in the range of 200–300 nm. CNCs with an aspect ratio of approximately 10–20 usually do not form network structure. Thus discrete emulsion droplets are stabilized by a layer of CNCs (Kalashnikova et al. 2013). Sphere-like CNC65-0.5 and CNC65-0.75 were also supposed to form a solid layer at oil–water interface. Previous study indicates Pickering emulsions stabilized by rod-like CNC I had larger emulsion ratio than those of rod-like CNC II (Li et al. 2018). Thus allomorph difference did not explain the above results. Sphere-like CNCs stabilized a larger volume of emulsion perhaps because at the same cellulose mass, they can cover more oil–water interface. Cellulose surface potential also played a role. Repulsive electrostatic force between droplets reduced their collision frequency and thus prevented their flocculation. Perhaps that is the reason why CNC65-0.75 performed better than CNC65-0.5.

Conclusions

CNCs with tunable shapes, sizes and allomorphs were successfully isolated from MCC by facile acidified zinc chloride treatment (ZnCl_2 : 55–65%, HCl : 0.5–1 M). The concentration of zinc chloride played an important role in monitoring the swelling level of MCC and the degree of hydrolysis of cellulose chains. With increase of ZnCl_2 concentration, the aspect ratio of CNCs decreased and cellulose allomorph transformed from native cellulose I to regenerated cellulose II. While rod-like CNC I (length: 181.3–279.3 nm) was produced in 55% ZnCl_2 , sphere-like CNC II (diameter: 25.8 to 292.3 nm) was obtained in 65% ZnCl_2 . A mixture of CNC I and CNC II with spherical, granular and rod-like shapes were observed in 60% ZnCl_2 . The CNCs exhibited high crystallinity of 49.6–68.4% and high thermal stability (T_{max} : 316.3–343.5 °C). Good stability of aqueous CNC suspensions could be observed especially for CNC65-0.75. Furthermore, CNCs prepared with this method were capable of forming oil-in-water Pickering emulsion, and that with spherical CNC65-0.75 exhibited larger emulsifying ratio and higher stability than that with commercialized CNCs. The spherical CNC II opens opportunities for the development of biomedicine, cosmetics, and food additives.

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Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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